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Designing functionalized polymers through post-polymerization modification of side chain leucine-based polymer

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ABSTRACT

The opportunity to modify the chemical functionality of the incorporated amino acid moiety in side chain amino acid-based polymers provides access to tailor their optical, mechanical, and biological properties. Methacrylate monomers from *C*-terminus modified amino acids and their subsequent polymerization leads to the formation of polymethacrylate polymers with pendant amine ($-NH_2$) groups. To this end, the post-polymerization reactions on the amine groups offer a facile route to prepare functionalized side chain amino acid-based polymeric materials. Herein, we synthesized side chain L-leucine pendant homopolymer using reversible addition-fragmentation chain transfer (RAFT) polymerization of Boc-L-leucine methacryloyloxyethyl ester (Boc-leu-HEMA) and subsequently the Boc group is deprotected to prepare $P(H_3N^+$-leu-HEMA). Post-polymerization modifications on the pendant amine groups were performed to incorporate fluorescent probe, propargyl group, methacrylic moiety, hydrophobic fatty acid, anionic pendant, and hydrophilic polyethylene glycol (PEG) units; which demonstrates an efficient platform to build a library of functionalized polymers from a single polymeric backbone. We have studied the solvatochromic behavior of fluorescence probe bearing polymers using fluorescence spectroscopy. The ability of the pendent methacrylic unit containing polymers to undergo crosslinking and the mechanical properties of the synthesized gel are investigated by the rheological study.

1. Introduction

Side chain amino acid-based polymers have received significant research interest as the presence of pendant amino acids imparts several intriguing characteristics into the polymeric materials including catalysis [1,2], optical properties [3], stimuli responsiveness [4], bioimaging [5], drug and gene delivery [6]. There are several methods to incorporate amino acid moieties into the polymeric architecture, such as polycondensation of amino acids, ring-opening polymerization of *N*-carboxy-α-amino acid anhydrides (NCAs) and controlled radical polymerization (CRP) of amino acid-functionalized monomers. Among them, CRP of amino acid-functionalized monomers is the most adaptable and potent method to prepare amino acid-based polymers [7]. Particularly, reversible addition-fragmentation chain transfer (RAFT) polymerization is widely used as it provides precise control over the molar mass and chain end functionality, with narrow dispersity (*Đ*) [8]. The diverse functional sites of amino acids allowed to attach the polymerizable moiety either at *C*-terminus or *N*-terminus or in some cases at the -*R* group (e.g., -SH or -OH). *C*-terminus modified monomers with unprotected *N*-terminus may not be polymerized through RAFT polymerization as the free amine group would interact with the RAFT agent [9]. Hence, protection of $-NH_2$ group of amino acid is crucial before RAFT polymerization, and their subsequent deprotection leads to the generation of side chain amino acid-based polymers with pendant free amine groups [10,2].

Direct polymerization of amino acid-based monomers with specialized functional moiety often suffers from intrinsic shortcomings due to the incompatibility of monomer functionality under polymerization conditions [11,12]. To supersede these hurdles, post-polymerization modification (PPM) of reactive macromolecular scaffolds was deemed to be a potent method for creating desired polymeric materials with multifunctional compositions [13,14]. PPM fundamentally relies on the direct polymerization or copolymerization of monomers containing chemoselective pendant groups, which are inert to polymerization conditions but can be activated under subsequent post-polymerization conditions, leading to the generation of a wide range of functional

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polymers. This method exhibits several advantages over the conventional direct polymerization of the related monomers: (1) abatement of unnecessary cost and complexity associated with the time-consuming functional monomers synthesis; (2) allowing the installation of functional groups that are conflicting with polymerization conditions; and (3) helps to fabricate a wide range of diverse functional polymers from a single macromolecular scaffold [15,16].

In the pursuit of an efficient route to coherent and controlled modification of polymers, different kinds of chemistries have been successfully implemented to the PPM reactions, which include Diels-Alder (DA) cycloaddition [17], Michael-type addition [18–20], and reaction with aldehydes, ketones [21], isocyanates, anhydrides, and activated esters [22]. Although several methods have been developed for PPM reactions, the use of catalysts for the reactions, and poor conversion due to the decreased reactivity of the functional group at polymeric scaffold restrict achieving their full potential for practical applications. Previous reports include post-polymerization modification of acrylic polymers containing Meldrum's acid [23]; copper-catalyzed multi-component reaction of terminal alkyne containing polymers [24]; sulfonyl azides and amines; modification of poly-L-lysine with fluorophores like dansyl group for protein sensing [25]; nucleophilic addition of alkyl amines to the styrenic carbodiimide [26]; and coupling of side chain alkoxyamine functionalized (R-O-NH_2) polymers with small aldehyde and ketone molecule [21].

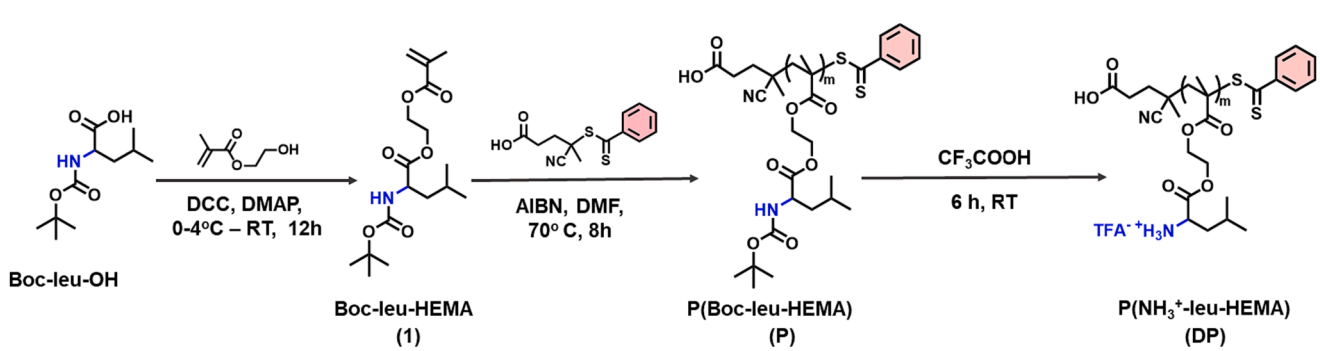
Since the preparation of different functionalized polymers from side chain amino acid-based single polymeric backbone has not been explored, herein we prepared a side chain leucine-based homopolymer, which provided a polymeric platform for an effective, specific, and reliable approach to prepare a functionalized polymeric library by performing several fascinating chemistries on the pendant amine (-NH_2) groups. A simple RAFT polymerization of Boc-L-leucine methacryloyloxyethyl ester (Boc-leu-HEMA) and subsequent deprotection of the Boc group was used to prepare leucine-based homopolymer with pendant primary amine groups. A series of different functionalized polymers were obtained from the side-chain leucine pendant homopolymer by performing modification of the amine group in leucine moiety with fluorescent probes, propargyl group, methacrylic unit, fatty acid, isocyanate, and *via* aza-Michael addition reactions with acrylate-based molecule.

2. Results and discussion

To functionalize the side chain amino acid-based polymer through the post polymerization modification of -NH_2 moieties, leucine pendant homopolymer was employed as a typical representative. The Boc-leu-HEMA methacrylate monomer (Scheme 1) and corresponding homopolymer P(Boc-leu-HEMA) were synthesized following the previous literature [27]. Typically, the Boc-leu-HEMA was polymerized with RAFT polymerization technique (Scheme 1) using 4-cyano-4-(thiobenzylthio)pentanoic acid (CTP) as a chain transfer agent and 2,2′-azobis (isobutyronitrile) (AIBN) as the initiator with the molar ratio of [Boc-leu-HEMA]/[CTP]/[AIBN] = 60:1:0.2 in *N*,*N*-dimethylformamide (DMF) at 70 °C. The homopolymer was characterized using ^{1}H NMR spectroscopy, and size exclusion chromatography (SEC). Characterization details of P(Boc-leu-HEMA) are summarized in Table S1. The absence of vinyl protons peaks of the Boc-leu-HEMA at 6.14 and 5.61 ppm (Fig. S1) in the ^{1}H NMR spectrum of P(Boc-leu-HEMA) (Fig. S2) suggested the successful removal of unreacted Boc-leu-HEMA and the purity of the homopolymer. Moreover, the aromatic proton peaks at 7.41–7.47, 7.75–7.85, and 7.86–7.95 ppm in the ^{1}H NMR spectrum of P(Boc-leu-HEMA) indicated the incorporation of the CTP as chain ends. The degree of polymerization (DP_n) of the homopolymer was calculated by comparing the chain-end signal at 7.41–7.47 ppm (two protons of the phenyl ring of CTP) with the protons peak of Boc-leu-HEMA at 4.55–3.75 ppm (five protons, -O-C$\boldsymbol{H_2}$-C$\boldsymbol{H_2}$-O- and -CO-C$\boldsymbol{H}$(NH-Boc) - of the side chain of the polymer). The estimated DP_n of the P(Boc-leu-HEMA) was found to be 58. The molar mass from NMR study ($M_{n,NMR}$) of P(Boc-leu-HEMA) was 20200 g/mol, which was calculated using following equation: $M_{n,NMR}$ = [DP_n × molecular weight (MW) of Boc-leu-HEMA + MW of CTP]. The formation of P(Boc-leu-HEMA) was also validated using SEC analysis (Fig. S3), where unimodal and narrow SEC trace was observed with the molar mass $M_{n,SEC}$ = 18800 g/mol ($Đ$ = 1.1).

The deprotection of the Boc groups of the P(Boc-leu-HEMA) polymer was performed by treating with trifluoroacetic acid (TFA) at room temperature for 6 h (Scheme 1). The formation of P(NH_3^+-leu-HEMA) (**DP**) was confirmed from the ^{1}H NMR and FTIR analysis (Figs. 1 and 2). The disappearance of -C(C$\boldsymbol{H_3}$)$_3$ protons peak at 1.45 ppm in **DP** and the appearance of a new peak at 8.64 ppm due to the pendant amine functionality corroborated the successful deprotection of Boc groups from the P(Boc-leu-HEMA). To record the FTIR spectrum of **DP**, we first neutralized the deprotected polymer (see Supporting Information). The presence of a broad peak at around 3380 cm^{-1} with a shoulder peak at 3300 cm^{-1} attributed to the N-H stretching mode ($\nu_{N\text{-}H}$) of -NH_2 groups in the polymer, further confirmed the formation of **DP**.

Next, the derivatization of the side chain -NH_2 groups of leucine pendant was carried out with different substrates to produce functional polymers **P1**-**P9** (Scheme 2). The synthesized polymers were characterized using ^{1}H NMR and FTIR spectroscopy. At the very beginning, to impart the fluorescence property into the polymer, we functionalized **DP** with 5-(dimethylamino)naphthalene-1-sulfonyl chloride (dansyl chloride) at room temperature in dichloromethane (DCM) using triethylamine (Et_3N) (Scheme S1), which exhibited fluorescence after the formation of sulfonamide adducts and would be useful for bioimaging and monitoring the actin polymerization [28]. The generation of dansyl tagged polymer (**P1**) was confirmed by the appearance of new aromatic protons peaks of the dansyl group in the range of 7.15–8.65 ppm and the -N$\boldsymbol{H}$-SO_2- proton peak overlapped within themselves (Fig. 1, P1). Following a similar strategy, the *p*-toluenesulfonyl group was incorporated in the **DP** *via* post-polymerization modification reaction with *p*-toluenesulfonyl chloride at room temperature using Et_3N in DCM, resulting the formation of **P2** (Scheme S2). Two distinct new peaks at



Scheme 1. Synthetic scheme of side chain leucine-based polymer.

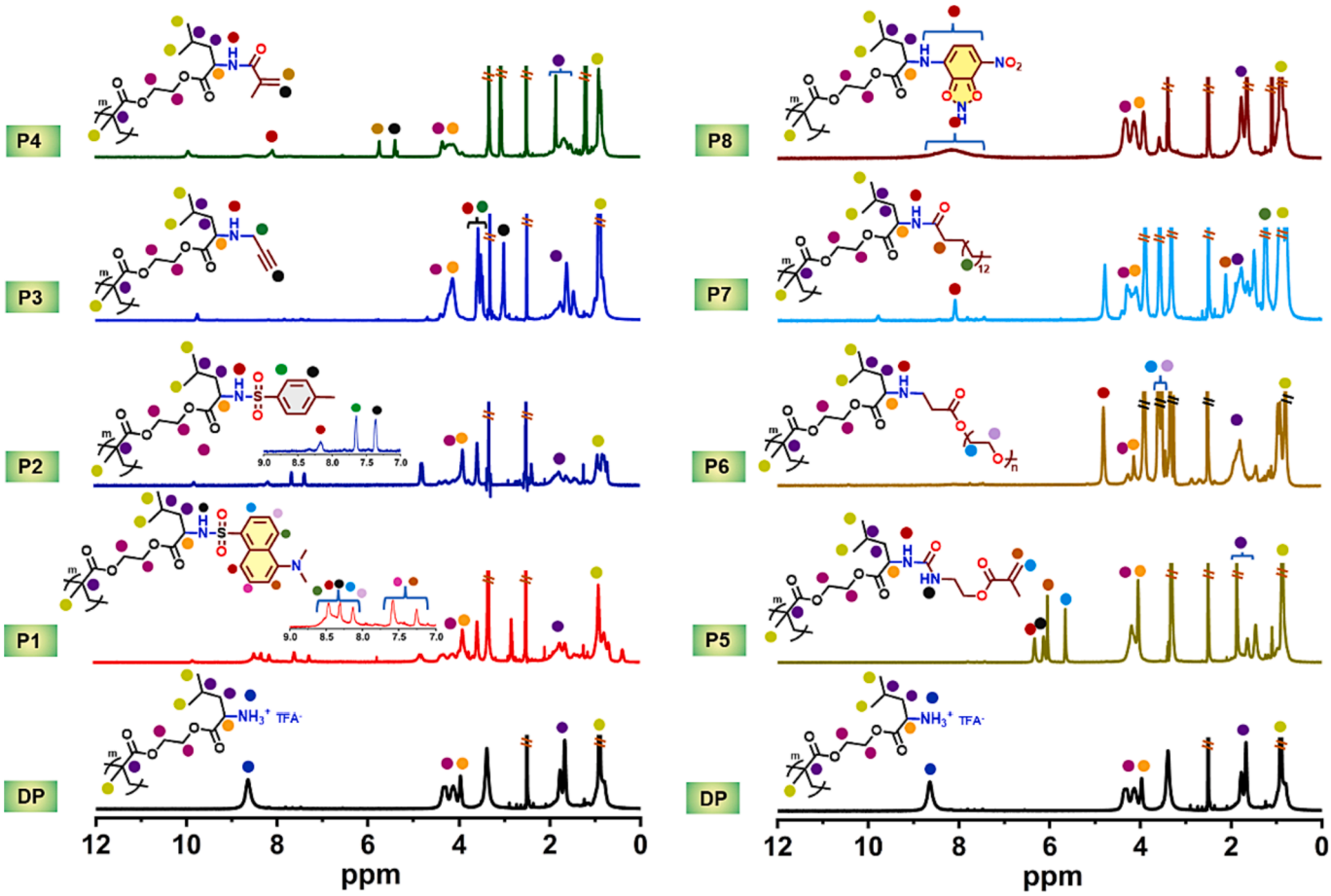


Fig. 1. ^{1}H NMR spectroscopic analysis of the functionalized polymers **P1**-**P8**, obtained from the post-polymerization modification of **DP**. The ^{1}H NMR spectra were recorded in DMSO-d_6. For better comparison, the **DP** spectrum is shown on both the left and right side of the figure.

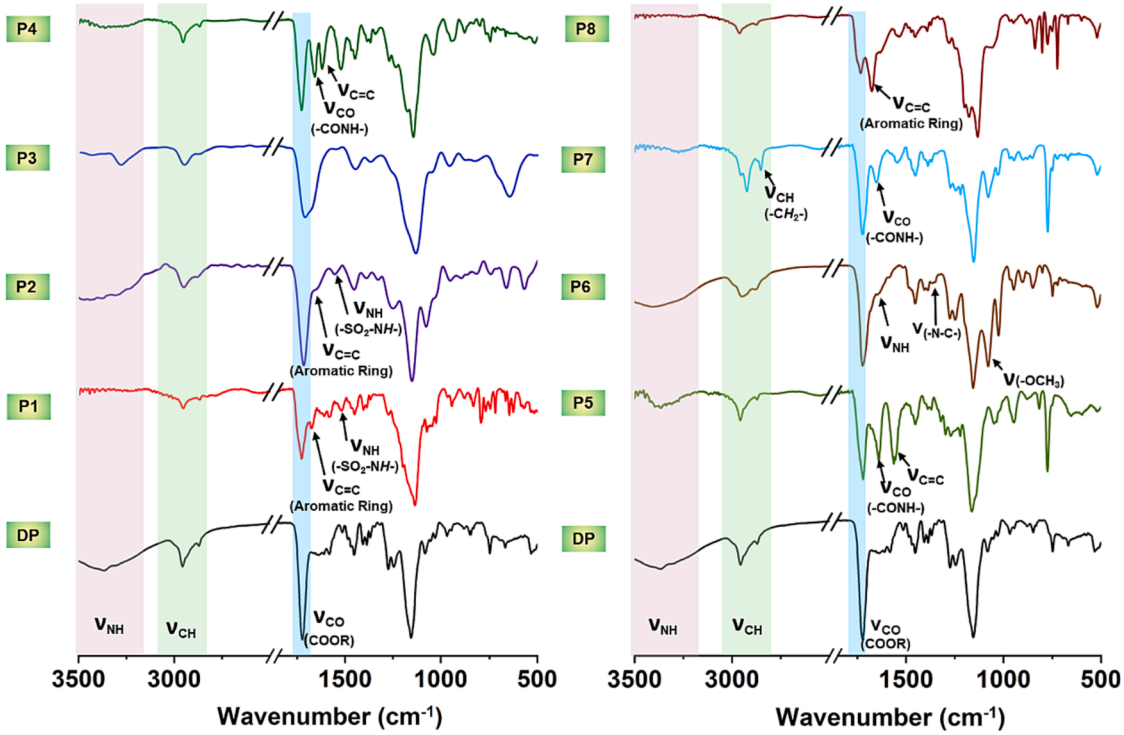
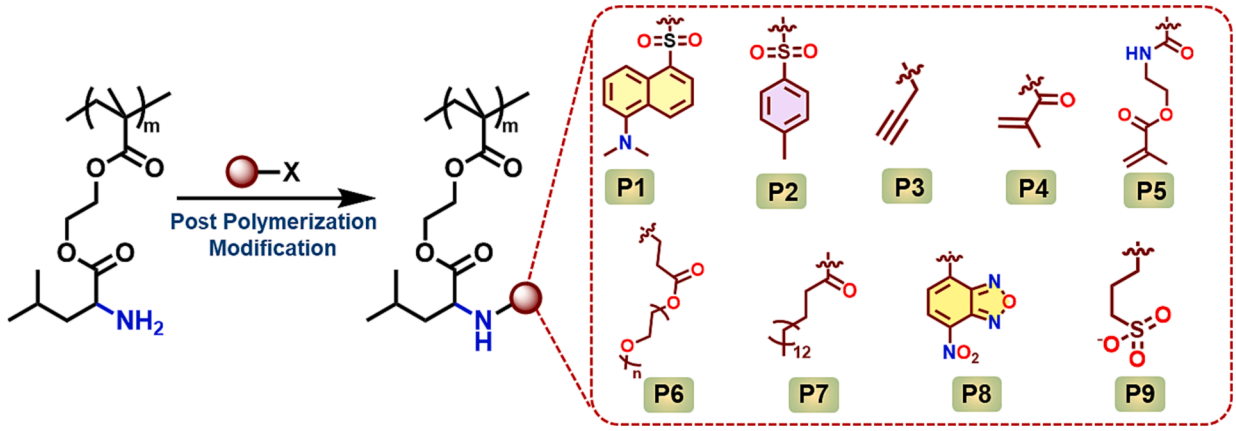


Fig. 2. FTIR spectroscopic characterization of **DP** and **P1**-**P8**. For better comparison, the **DP** spectrum is shown on both the left and right side of the figure.



Scheme 2. Schematic representation for the synthesis of a library of functionalized polymers through post-polymerization modification of side chain leucine-based homopolymer.

7.65 and 7.36 ppm due to the aromatic protons of *p*-toluenesulfonyl group and a peak at 8.17 ppm of -N***H***-SO_2- in the ^{1}H NMR spectrum (Fig. 1, P2) implied the successful functionalization of the polymer. We envisaged that this modification could potentially be utilized not only for biomedical applications but also for energy storage or battery applications [29].

The incorporation of the propargyl group into the polymeric scaffold is crucial as it undergoes a "click reaction", which is an efficient tool to conjugate a wide range of functional moieties into the polymer [30]. Thus, the propargyl group was introduced in **DP** with the reaction of propargyl bromide at room temperature in DMF using potassium carbonate as a base (Scheme S3). The propargyl-modified polymer (**P3**, Scheme 2) exhibited a new distinguished resonance signal at 3.42–3.65 ppm for (-N***H***-CH_2- and -C***H***$_2$-C≡CH), and at 3.01 ppm for propargyl proton (-C≡C***H***) (Fig. 1, P3). Of note, there is a possibility that double functionalization happened in the -NH_2 group of **DP**, which cannot be controlled under that reaction condition. The methacrylic unit conjugated polymer, **P4** was also synthesized from **DP** *via* post-polymerization modification with methacrylic chloride in DCM using Et_3N (Scheme S4). This polymer is advantageous for the fabrication of covalently cross-linked three-dimensional materials [31]. The **P4** was primarily characterized using ^{1}H NMR spectroscopy, where it showed new resonance signals at 5.72 and 5.38 ppm due to the vinyl protons (- $(CH_3)C = CH_2$) of the methacrylic unit (Fig. 1, P4). Furthermore, the appearance of a new amide (-N***H***-CO-) peak at 8.09 ppm suggested the successful incorporation of the methacrylic group. We also manifested that the methacrylic unit can be incorporated into the polymeric system through the reaction of **DP** and 2-isocyanatoethyl methacrylate in DCM using Et_3N (Scheme S5). The functionalized polymer, **P5** showed two distinct peaks for vinyl protons (- $(CH_3)C = CH_2$) of the methacrylate unit at 5.66 and 6.05 ppm, whereas peaks for -N***H***-CO-NH- and -NH-CO-N***H***- arise respectively at 6.33 and 6.13 ppm, pointed to the successful formation of **P5** (Fig. 1, P5).

Next, an aza-Michael addition reaction was performed on **DP** with polyethylene glycol methyl ether acrylate (PEGA) at 50 °C for 24 h (Scheme S6) that leads to the formation of product **P6**. The appearance of new peaks at 4.80 ppm due to the -N***H***-CH_2- and at 3.46–3.64 ppm from the -O-C***H***$_2$-C***H***$_2$-O- protons (from the incorporated PEGA unit) indicated the successful formation of aza-Michael adduct (Fig. 1, P6). Palmitic acid, a long-chain fatty acid was also successfully coupled with the **DP** to introduce hydrophobicity into the polymer chain using an amide coupling reaction with 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) and hydroxybenzotriazole (HOBt) at room temperature for 12 h (Scheme S7). The successful incorporation of palmitate moiety in the polymer (**P7**) side chain was confirmed by the appearance of new peaks at 8.09 (-N***H***-CO-), 2.13 (-CO-C***H***$_2$-), and 1.24 ppm (-CH_2- (C***H***$_2$)$_{12}$-CH_3) in the ^{1}H NMR spectrum (Fig. 1, P7). Another fluorescent probe 4-chloro-7-nitrobenzofurazan (NBD-Cl) was also incorporated *via* post-polymerization modification using activated nucleophilic substitution between the reaction of NBD-Cl and amine group of **DP** in anhydrous methanol (MeOH) at room temperature for 24 h (Scheme S8). The ^{1}H NMR spectroscopic analysis evidenced the successful synthesis of NBD-based fluorescent polymer (**P8**), where a broad peak near 8.22 ppm (Fig. 1, P8) was observed on account of aromatic proton of the nitrobenzofuran ring, and the aromatic secondary amine proton. Additionally, in all the functionalization reactions, the disappearance of the free -NH_2 of **DP** peaks at 8.64 ppm confirmed the successful derivatization of the amine group. Post-polymerization modification of **DP** with 1,3-propane sultone was performed in acetonitrile at 82 °C for 12 h (Scheme S9). ^{1}H NMR spectrum of **P9** exhibited new peaks at 2.96, 2.63, and 1.96 ppm as shown in Fig. S4, indicating successful modification of **DP** with anionic moiety. Additionally, the zeta potential analysis was conducted using **P9** in de-ionized water, revealing values of −2.2 ± 1.2, −24.8 ± 1.9, and −38.8 ± 1.5 mV at pH 5, 7, and 9, respectively. Thus, **P9** exhibits anionic characteristics in aqueous solution.

To calculate the percentage modification of the functionalized polymers **P1**-**P9** with respect to the side chain leucine unit, we considered the protons signals of **DP** at 4.55–3.75 ppm (due to 5 protons; 4 protons of -O-C***H***$_2$-C***H***$_2$-O- and one chiral proton of side chain leucine) as standard. The distinguishable proton signals of each functionalized polymer were compared to the standard to evaluate the percentage modification of **P1**-**P9**. The targeted modification percentage, the actual percentage of modification value, $M_{n,SEC}$ and $Đ$ (Fig. S5) obtained from the ^{1}H NMR spectroscopic study and SEC study are summarized in Table 1. Moreover, the peak at ~ 300 nm in the absorption spectra of the functionalized polymers confirmed the presence of chain end thiocarbonylthio groups in the polymers (Fig. S6).

For the **P1**, we targeted a modification of 25% but the ^{1}H NMR analysis suggested 23% modification. Here, the percentage of modification was calculated by comparing the integration value of the ^{1}H NMR signal of the dansyl unit at 7.25 ppm (due to a single aromatic proton peak, (Fig. 1, P1)) with the peaks at 4.55–3.75 ppm due to 5 protons from the polymer side chain (Fig. 1, **DP**). We have calibrated the integration value of 4.55–3.75 ppm peak as 5, and we calculated the integration value for a single aromatic proton peak at 7.25 ppm of newly installed dansyl unit as 0.23, which indicated the modification of 23%. Of note, if the modification was 100% then we should get the integration value for the single proton peak at 7.25 ppm peak as 1. Likewise, we have calculated the percentage of modification of **P2**, **P4**, **P5**, and **P7**. In case of **P2**, the peak at 7.36 ppm (due to a two aromatic proton) was used to calculate the modification percentage and the obtained value is 29% (target ~ 50%). In a similar way, the standard peak is compared with the

Table 1
Percentage of modification upon completion of post-polymerization reactions.

Polymer	Reagent used for $-NH_2$ derivatization	% modification[a] (Observed)	% modification (Targeted)	Yield[b] (%)	$M_{n,SEC}^{d}$ (g/mol)	$Đ^{d}$
DP	–	–	–	–	4900	1.13
P1	Dansyl cholride	23	25	80	5100	1.15
P2	4-Methylbenzenesulfonyl chloride	29	50	65	9500	1.18
P3	Propargyl bromide	~95[c]	100	72	11,100	1.28
P4	Methacryloyl chloride	54	80	85	N.D.	N.D.
P5	2-Isocyanatoethyl methacrylate	70	80	82	N.D.	N.D.
P6	PEGA	–	100	70	N.D.	N.D.
P7	Palmitic acid	40	50	60	9500	1.18
P8	NBD-Cl	–	25	88	7500	1.16
P9	1,3-Propane sultone	–	100	85	N.D.	N.D.

[a]Modification % values were determined using ^{1}H NMR spectroscopic analysis by comparing the integration value of the signal at 4.55–3.75 ppm from the polymer side-chain protons with respect to the integration value of the peak from incorporated functionalized unit. [b]Calculated gravimetrically after purification. [c]Modification % of the mixture of single and double functionalized product. [d]Measured from SEC analysis in DMF.

single vinyl proton (-$(CH_3)C = C\boldsymbol{H_2}$) peaks at 5.72 and 5.66 ppm due to **P4** and **P5**, resulting the percentage of modification as 54 and 70 respectively, which was targeted as 80% in each cases. For **P7**, the resonance peak at 8.09 ppm (due to -N***H***-CO-) was used to calculate the percentage of modification as 40% (target ~ 50%) (Table 1). For **P3**, the peak at 3.01 ppm (due to the single protons of -CH_2-C≡C***H***) is used for the calculation, which led to the percentage of modification value as ~ 95% (target ~ 100%). We could not determine the percentage modification for the **P6**, **P8**, and **P9** with ^{1}H NMR analysis as the peaks due to the newly incorporated functionalized units interfering with the standard peak at 4.33–3.90 ppm obtained from the polymer side chain. Thus, we performed the fluorescamine assay for **P6** and **P9** to qualitatively determine the percentage of free -NH_2 group in the polymers. As shown in the Fig. S7, the free -NH_2 content in **P6** and **P9** were 25% and 12%, respectively, whereas, the free -NH_2 content in **DP** was considered 100%. Nevertheless, these results indicate an efficient conversion of the -NH_2 derivatization reaction in a macromolecular motif with adequate yield.

The **P1**-**P8** polymers were further characterized using FTIR spectroscopy (Fig. 2). The presence of a strong band near 1730 cm^{-1} (due to the -C = O stretching mode) and an intense broad band with a small hump at ~ 3300 cm^{-1} (due to -NH stretching mode) attributed to the ester and -NH_2 functionality of **DP**, respectively. The **P1**-**P8** exhibited a broad band of reduced intensity at 3300 cm^{-1} due to $\nu_{N\text{-}H}$ of the secondary amine group, suggested the successful derivatization reaction on the pendant -NH_2 group.

In case of **P1** and **P2**, a new band near 1520–1550 cm^{-1} appeared due to the $\nu_{N\text{-}H}$ bending of the -SO_2-NH- functionality [32]. The formation of **P3** was confirmed from the generation of a characteristics -C≡C- stretching frequency at 2130 cm^{-1} (Fig. S8) [33]. A new strong peak at 1640–1660 cm^{-1} was observed due to the -C = O stretching mode of amide -CO-NH- functionality present in **P4**, **P5**, and **P7** polymers [34], which was absent in the case of **DP**, thus indicating the successful post-polymerization modifications in **P4**, **P5**, and **P7**. The strong band at 1620 cm^{-1} for **P4** (-C = C- stretching) and 1570 cm^{-1} for **P5** (-C = C- stretching) suggested the presence of pendant alkene moiety into the polymer chain. However, this shifting of -C = C- stretching to lower wavenumber in case of **P4** and **P5** is observed due to the conjugation with -C = O. In the case of **P6**, a band at 1510 cm^{-1} arose due to $\nu_{N\text{-}H}$ bending of secondary amine (-NH-CH_2-). Also, the peaks at 1350 cm^{-1} (due to -N-C- stretching) [32] and 1075 cm^{-1} (because of -O-CH_3 stretching) further proved the successful synthesis of **P6** *via* aza-Michael addition reaction of **DP** with PEGA. In case of **P1, P2** and **P8** the -C = C- stretching of the incorporated aromatic ring arose at 1650-1670 cm^{-1} [32]. Furthermore, the generation of bands at 2800–3000 cm^{-1} ascribed to the -CH stretching mode, and particularly in case of **P7** incorporation of a long alkyl chain led to the generation of the comparatively more intense band at 2850 cm^{-1} due to the $\nu_{C\text{-}H}$ stretching band of -CH_2-.

After successful synthesis of a library of functionalized polymers, we focused to investigate their interesting properties. Given the presence of fluorescent probes like dansyl group in **P1** and the NBD-group in **P8** (Fig. 3A and 3B), we investigated their fluorescent behavior in different solvents like water, MeOH, and dimethylsulfoxide (DMSO). Of note, other non-polar solvents were not chosen because **P1** and **P8** were poorly soluble in those solvents. The **P1** exhibited an intense absorption band at 330 nm (Fig. S9) because of the intramolecular charge transfer of the aminonaphthalene-sulfonate moiety [35,36]. Similar absorption spectra of **P1** in different solvents (Fig. S9) suggested that the solvent polarity has less effect on the ground state electronic transition of the chromophore. In contrast, upon changing the solvent from DMSO to water the emission maximum (excitation wavelength, λ_{ex} = 330 nm) shifted from 450 nm to 505 nm (Fig. 3C). This bathochromic shift of emission maximum with increasing solvent polarity suggested the stabilization of polar emitting excited states in polar solvents [37]. Moreover, the concentration dependent emission spectra of **P1** were also recorded in each solvent (Fig. S10). The absorption and emission spectra of **P8** were also investigated (Fig. S11 and Fig. S12). The absorption spectra depicted two characteristics peaks at around 332 nm (due to π-π^* transition) and 470 nm (for the charge transfer band) [38]. When the polymer was excited at 470 nm wavelength, fluorescence emission maximum was observed at 530 nm (Figs. 3D and S9). However, a little bathochromic shift of emission maximum is observed with increasing the solvent polarity from MeOH to water (Fig. 3D) [39]. This aqueous soluble polymer with a significant fluorescent signature validates its potential in the field of chemosensor [40], imaging [41], sensing or labelling purpose [42], and molecular recognition [43].

Next, the gelation properties of the methacrylic unit-modified polymer were investigated. Although both the **P4** and **P5** could be used for this study, we selected **P4** for the proof of concept. The methacrylate unit conjugated polymer **P4** underwent crosslinking in the presence of AIBN, leading to the formation of a gel (Scheme S10 and Fig. S13). The FTIR spectroscopic (Fig. S14) characterization of the synthesized gel indicated a $\nu_{N\text{-}H}$ stretching band at 3450 cm^{-1}, -C = O stretching of all the other functional moiety is observed ca. 1650 cm^{-1}. Rheological measurements were performed to investigate the mechanical strength of the synthesized gel (Fig. 4). In order to determine the linear viscoelastic regime, storage modulus (G') and loss modulus (G'') *versus* percentage of strain measurements was carried out, which suggested a linear viscoelastic regime up to 10% of shear strain (Fig. 4A). Therefore, the frequency (ω) sweep experiment, which monitored the change in G' and G'' as a function of applied frequency has been carried out at a constant strain of 1% (Fig. 4B). A plateau of G' is observed over the frequency range of 0.01–10.00 Hz, and the G' value is obtained as ~ 2.5×10^6 Pa. The significantly higher value of G' than the G'' in all the frequency ranges indicated the solid-like nature of the gel with robust mechanical properties [44,45].

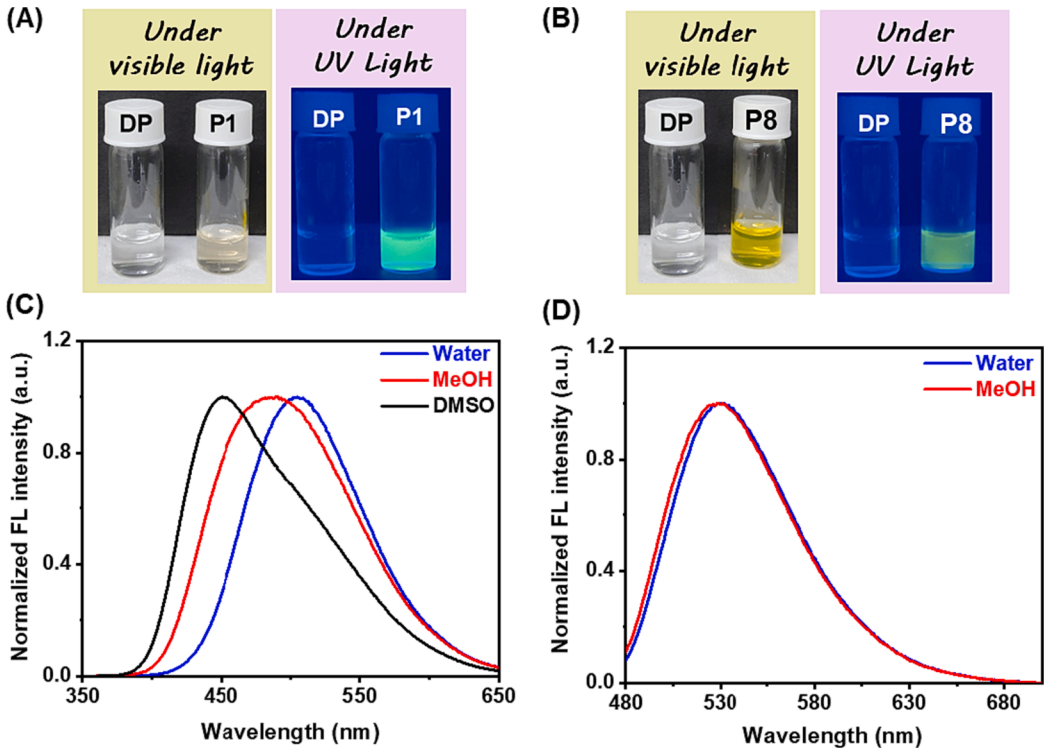


Fig. 3. Photographic representation of (A) **DP**, **P1**, and (B) **DP**, **P8** under visible light and UV light (365 nm). Emission spectra of (C) **P1** (λ_{ex} = 330 nm) and (D) **P8** (λ_{ex} = 470 nm) in different solvents.

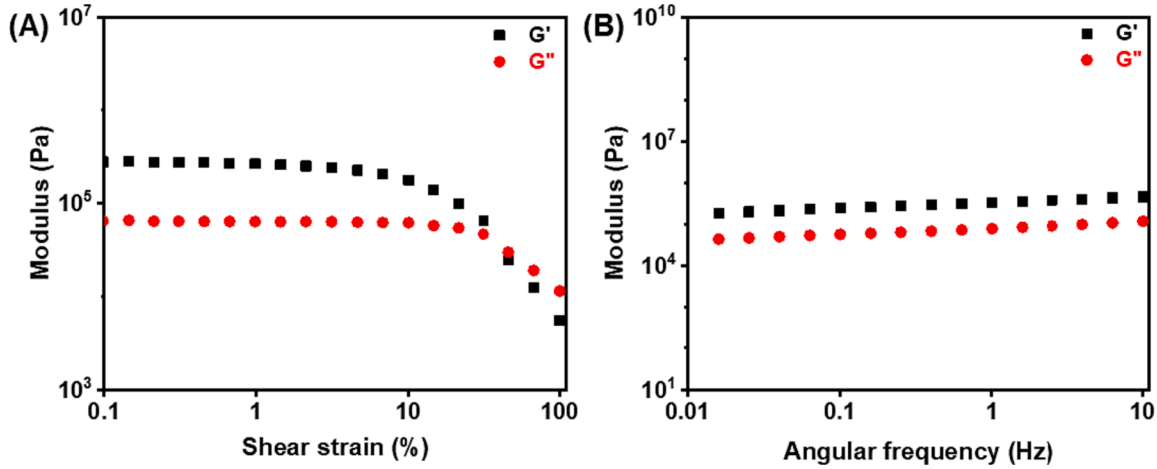


Fig. 4. (A) Storage modulus (G') and loss modulus (G'') on strain sweep experiment of the synthesized hydrogel. (B) Frequency dependence of storage modulus (G') and loss modulus (G'') at a constant strain rate of 1%.

3. Conclusions

In summary, this report portrays the synthesis of a diverse range of functional polymers from side chain leucine-containing polymer *via* post-polymerization modification reactions. The coupling of fluorophore (e.g., dansyl and NBD group), *p*-toluenesulfonyl group, methacrylic moiety, alkyne group, hydrophilic PEG, and long alkyl chain based hydrophobic unit with the pendant -NH_2 groups of **DP** was successfully performed under mild reaction conditions, leading to the generation of a library of functionalized polymers. The dansyl and NBD-modified polymers (**P1** and **P8**) exhibit fluorescent properties and we envisage that these modifications would be useful for biomedical applications and cell imaging. Interestingly, the inherent secondary amine functionality present in the synthesized sulfonamide-based polymer **P2** can further undergo aza-Michael reaction to generate additional functionality; for example, coupling with electron-deficient alkenes will help to prepare polymeric β-amino acids [46]. Given the presence of the pendant alkyne groups in **P3**, it holds great promise for undergoing conjugation with biomolecules [47], fluorescent probe [48], nanomaterials [49], or other surfaces containing azide units through "click reaction" [50,51]. The methacrylate-modified polymers **P4** and **P5** undergo crosslinking and the rheological measurement of the synthesized cross-linked gel from **P4** revealed its robust mechanical property. The hydrophilic PEG segment *via* aza-Michael reaction (**P6**), anionic moiety (**P9**), and hydrophobic fatty acid (palmitate in **P7**) were successfully introduced into the **DP**, and interesting self-assembly characteristics in aqueous medium are expected [52], with these polymers with possible therapeutic applications to treat different amyloidogenic disorders [53].

Taken together, a library of functionalized polymers from a single polymeric backbone with side-chain amino acid pendants bearing free -NH_2 functionalities have a bright prospect for diverse applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112445.

References

[1] E.R.L. Brisson, Z. Xiao, L.A. Connal, Amino acid functional polymers: biomimetic polymer design enabling catalysis, chiral materials, and drug delivery, Australian Journal of Chemistry 69 (2016) 705–716, https://doi.org/10.1071/CH16028.
[2] T. Maity, S. Paul, P. De, Side-chain amino acid-based macromolecular architectures, J. Macromol. Sci., Part A, Pure and Applied Chemistry 60 (2023) 2–17, https://doi.org/10.1080/10601325.2023.2169158.
[3] K. Bauri, S. Pant, S.G. Roy, P. De, Dual pH and temperature responsive helical copolymer libraries with pendant chiral leucine moieties, Polymer Chemistry 4 (2013) 4052–4060, https://doi.org/10.1039/C3PY00434A.
[4] K. Shoji, M. Nakayama, T. Koseki, K. Nakabayashi, H. Mori, Threonine-based chiral homopolymers with multi-stimuli-responsive property by RAFT polymerization, Polymer 97 (2016) 20–30, https://doi.org/10.1016/j.polymer.2016.05.003.
[5] N. Choudhury, B. Ruidas, B. Saha, K. Srikanth, C.D. Mukhopadhyay, P. De, Multifunctional tryptophan-based fluorescent polymeric probes for sensing, bioimaging and removal of Cu2+ and Hg2+ ions, Polymer Chemistry 11 (2020) 2015–2026, https://doi.org/10.1039/C9PY01892A.
[6] S.S. Gupta, V. Mishra, M.D. Mukherjee, P. Saini, K.R. Ranjan, Amino acid derived biopolymers: recent advances and biomedical applications, International Journal of Biological Macromolecules 188 (2021) 542–567, https://doi.org/10.1016/j.ijbiomac.2021.08.036.
[7] R.K. O'Reilly, Using controlled radical polymerization techniques for the synthesis of functional polymers containing amino acid moieties, Polymer International 59 (2010) 568–573, https://doi.org/10.1002/pi.2830.
[8] K. Bauri, S.G. Roy, P. De, Side-chain amino-acid-derived cationic chiral polymers by controlled radical polymerization, Macromolecular Chemistry and Physics 217 (2016) 365–379, https://doi.org/10.1002/macp.201500271.
[9] M.R. Hill, R.N. Carmean, B.S. Sumerlin, Expanding the scope of RAFT polymerization: recent advances and new horizons, Macromolecules 48 (2015) 5459–5469, https://doi.org/10.1021/acs.macromol.5b00342.
[10] H. Sun, F. Meng, A.A. Dias, M. Hendriks, J. Feijen, Z. Zhong, α-Amino acid containing degradable polymers as functional biomaterials: rational design, synthetic pathway, and biomedical applications, Biomacromolecules 12 (2011) 1937–1955, https://doi.org/10.1021/bm200043u.
[11] T. Kubo, G.M. Scheutz, T.S. Lattya, B.S. Sumerlin, Synthesis of functional and boronic acid-containing aliphatic polyesters *via* Suzuki coupling, Chemical Communications 55 (2019) 5655–5658, https://doi.org/10.1039/C9CC01975H.
[12] S. Reinicke, P. Espeel, M.M. Stamenović, F.E. Du Prez, One-pot double modification of P(NIPAAM): a tool for designing tailor-made multiresponsive polymers, ACS Macro Letters 2 (2013) 539–543, https://doi.org/10.1021/mz4002222.
[13] W.A. Cunningham, Sulfur. III, Journal of Chemical Education 12 (1935) 120–124, https://doi.org/10.1021/ed012p120.
[14] M.A. Gauthier, M.I. Gibson, H.A. Klok, Synthesis of functional polymers by post-polymerization modification, Angewandte Chemie 48 (2008) 48–58, https://doi.org/10.1002/anie.200801951.
[15] K.A. Günay, P. Theato, H.A. Klok, Standing on the shoulders of hermann staudinger: post-polymerization modification from past to present, Journal of Polymer Science, Part A: Polymer Chemistry 51 (2013) 1–28, https://doi.org/10.1002/pola.26333.
[16] J. Romulus, J.T. Henssler, M. Weck, Postpolymerization modification of block copolymers, Macromolecules 47 (2014) 5437–5449, https://doi.org/10.1021/ma5009918.
[17] D. Zhang, L. Pan, Y. Li, B. Wang, Y. Li, Synthesis and the reaction of anthracene-containing polypropylene: a promising strategy for facile, efficient functionalization of isotactic polypropylene, Macromolecules 50 (2017) 2276–2283, https://doi.org/10.1021/acs.macromol.6b02550.
[18] P. Li, S.R. De, B.S. Gondi, Sumerlin, End group transformations of raft-generated polymers with bismaleimides: functional telechelics and modular block copolymers, Journal of Polymer Science, Part A: Polymer Chemistry 46 (2008) 5093–5100, https://doi.org/10.1002/pola.22837.
[19] B. Cengiz, N. Ejderyan, A. Sanyal, Functional polymeric coatings: Thiol-maleimide 'click' chemistry as a powerful surface functionalization tool, J. Macromol. Sci., Part A, Pure and Applied Chemistry 59 (2022) 443–455, https://doi.org/10.1080/10601325.2022.2071725.
[20] E.J. Park, T.N. Gevrek, R. Sanyal, A. Sanyal, Indispensable platforms for bioimmobilization: Maleimide-based thiol reactive hydrogels, Bioconjugate Chemistry 25 (2014) 2004–2011, https://doi.org/10.1021/bc500375r.
[21] M.R. Hill, S. Mukherjee, P.J. Costanzo, B.S. Sumerlin, Modular oxime functionalization of well-defined alkoxyamine-containing polymers, Polymer Chemistry 3 (2012) 1758–1762, https://doi.org/10.1039/c1py00451d.
[22] A. Das, P. Theato, Activated ester containing polymers: opportunities and challenges for the design of functional macromolecules, Chemical Reviews 116 (2016) 1434–1495, https://doi.org/10.1021/acs.chemrev.5b00291.
[23] L. Yuan, L. He, Y. Wang, X. Lang, F. Yang, Y. Zhao, H. Zhao, Two-and three-component post-polymerization modifications based on Meldrum's acid, Macromolecules 53 (2020) 3175–3181, https://doi.org/10.1021/acs.macromol.0c00482.
[24] R. Kakuchi, P. Theato, Three-component reactions for post-polymerization modifications, ACS Macro Letters 2 (2013) 419–422, https://doi.org/10.1021/mz400144q.
[25] S. Tomita, S. Ishihara, R. Kurita, Environment-sensitive turn-on fluorescent polyamino acid: Fingerprinting protein populations with post-translational modifications, ACS Applied Materials & Interfaces 9 (2017) 22970–22976, https://doi.org/10.1021/acsami.7b05360.
[26] H.E. Houck, K.A. McConnell, C.J. Klingler, A.L. Koenig, G.K. Himka, M.B. Larsen, Postpolymerization modification by nucleophilic addition to styrenic carbodiimides, ACS Macro Letters 12 (2023) 1112–1117, https://doi.org/10.1021/acsmacrolett.3c00382.
[27] K. Bauri, S. Ghosh Roy, S. Pant, P. De, Controlled synthesis of amino acid-based pH-responsive chiral polymers and self-assembly of their block copolymers, Langmuir 29 (2013) 2764–2774, https://doi.org/10.1021/la304918s.
[28] B. Maiti, P. Dutta, S. Seal, S. Pal, P. De, S. Maiti, Side-chain amino acid based cationic polymer induced actin polymerization, Journal of Materials Chemistry B 5 (2017) 1218–1226, https://doi.org/10.1039/C6TB02814D.
[29] S. Feng, M. Huang, J.R. Lamb, W. Zhang, R. Tatara, Y. Zhang, Y.G. Zhu, C. F. Perkinson, J.A. Johnson, Y. Shao-Horn, Molecular design of stable sulfamide-and sulfonamide-based electrolytes for aprotic Li-O_2 batteries, Chem 5 (2019) 2630–2641, https://doi.org/10.1016/j.chempr.2019.07.003.
[30] Y. Shi, X. Caoa, H. Gao, The use of azide–alkyne click chemistry in recent syntheses and applications of polytriazole-based nanostructured polymers, Nanoscale 8 (2016) 4864–4881, https://doi.org/10.1039/C5NR09122E.
[31] E. Markovic, S. Clarke, J. Matisons, G.P. Simon, Synthesis of POSS− methyl methacrylate-based cross-linked hybrid materials, Macromolecules 41 (2008) 1685–1692, https://doi.org/10.1021/ma702103z.
[32] D.L. Pavia, G.M. Lampman, G.S. Kriz, J.A. Vyvyan, Introduction to spectroscopy, fifth ed., Cengage Learning, 2015.
[33] M.E. Pekdemir, A. Tukur, M. Coskun, Thermal and dielectric investigation of magnetic nanoparticles functionalized with PVC via click chemistry, Polymer Bulletin 78 (2021) 6047–6057, https://doi.org/10.1007/s00289-020-03404-z.
[34] Y. Bernhard, J.F. Van Guyse, M. Purino, R. Hoogenboom, Direct synthesis of poly (*N*-alkyl acrylamide)(co) polymers with pendant reactive amino groups by organocatalyzed amidation of polymethylacrylate, Eur. Polym. J. 192 (2023) 112077 (1-11). DOI: 10.1016/j.eurpolymj.2023.112077.
[35] S. Mocanu, G. Ionita, I. Matei, Solvatochromic characteristics of dansyl molecular probes bearing alkyl diamine chains, Spectrochim. Acta A Mol. Biomol. Spectrosc. 237 (2020) 118413 (1-8). DOI: 10.1016/j.saa.2020.118413.
[36] C.J. Seliskar, L. Brand, Electronic spectra of 2-aminonaphthalene-6-sulfonate and related molecules. II. Effects of the solvent medium on the absorption and fluorescence spectra, Journal of the American Chemical Society 93 (1971) 5414–5420, https://doi.org/10.1021/ja00750a017.
[37] C. Katan, F. Terenziani, O. Mongin, M.H. Werts, L. Porres, T. Pons, J. Mertz, S. Tretiak, M. Blanchard-Desce, Effects of (multi) branching of dipolar chromophores on photophysical properties and two-photon absorption, The Journal of Physical Chemistry. A 109 (2005) 3024–3037, https://doi.org/10.1021/jp044193e.
[38] A. Chattopadhyay, Chemistry and biology of N-(7-nitrobenz-2-oxa-1, 3-diazol-4-yl)-labeled lipids: fluorescent probes of biological and model membranes, Chemistry and Physics of Lipids 53 (1990) 1–15, https://doi.org/10.1016/0009-3084(90)90128-E.
[39] T. Dutta, K. Pal, A.L. Koner, Cellular metabolic activity marker *via* selective turn-on detection of transporter protein using nitrobenzoxadiazole-based fluorescent reporter. Sci Rep 10 (2020) 4166 (1-9). DOI: 10.1038/s41598-020-60954-y.
[40] S. Kumar, P. De, Fluorescent labeled dual-stimuli (pH/thermo) responsive self-assembled side-chain amino acid-based polymers, Polymer 55 (2014) 824–832, https://doi.org/10.1016/j.polymer.2013.12.054.

[41] R.F. de Almeida, T.C. Santos, L.C. da Silva, J. Suchodolski, A. Krasowska, K. Stokowa-Sołtys, M. Puchalska, R. Starosta, NBD-derived diphenyl(aminomethyl) phosphane – A new fluorescent dye for imaging of low pH regions and lipid membranes in living cells, Dyes Pigm. 184 (2021) 108771 (1-14). DOI: 10.1016/j.dyepig.2020.108771.
[42] L.P. Datta, D. De, U. Ghosh, T.K. Das, RAFT-derived fatty acid-based stimuli responsive fluorescent block copolymers as DNA sensor and cargo delivery agent, Polymer 138 (2018) 103–112, https://doi.org/10.1016/j.polymer.2018.01.044.
[43] S. Medel, P. Bosch, C. de la Torre, P. Ramírez, Click chemistry to fluorescent hyperbranched polymers. 1–Synthesis, characterization and spectroscopic properties, European Polymer Journal 59 (2014) 290–301, https://doi.org/10.1016/j.eurpolymj.2014.08.008.
[44] R. Laishram, S. Sarkar, I. Seth, N. Khatun, V.K. Aswal, U. Maitra, S.J. George, Secondary nucleation-triggered physical cross-links and tunable stiffness in seeded supramolecular hydrogels, Journal of the American Chemical Society 144 (2022) 11306–11315, https://doi.org/10.1021/jacs.2c03230.
[45] S.G. Roy, A. Kumar, P. De, Amino acid containing cross-linked co-polymer gels: pH, thermo and salt responsiveness, Polymer 85 (2016) 1–9, https://doi.org/10.1016/j.polymer.2016.01.021.
[46] M. Ashfaq, R. Tabassum, M.M. Ahmad, N.A. Hassan, H. Oku, G. Rivera, Enantioselective synthesis of β-amino acids: A Review, Medicinal Chemistry 5 (2015) 295–309, https://doi.org/10.4172/2161-0444.1000278.
[47] Q. Wang, T.R. Chan, R. Hilgraf, V.V. Fokin, K.B. Sharpless, M.G. Finn, Bioconjugation by copper (I)-catalyzed azide-alkyne [3+ 2] cycloaddition, Journal of the American Chemical Society 125 (2003) 3192–3193, https://doi.org/10.1021/ja021381e.
[48] T. Ishizuka, H. Liu, K. Ito, Y. Xu, Fluorescence imaging of chromosomal DNA using click chemistry, Scientific Reports 6 (2016) 1–10, https://doi.org/10.1038/srep33217.
[49] X. Liu, P. Gong, P. Song, F. Xie, A.L. Miller II, S. Chen, L. Lu, Rapid conjugation of nanoparticles, proteins and siRNAs to microbubbles by strain-promoted click chemistry for ultrasound imaging and drug delivery, Polymer Chemistry 10 (2019) 705–717, https://doi.org/10.1039/C8PY01721B.
[50] E.M. Sletten, C.R. Bertozzi, From mechanism to mouse: a tale of two bioorthogonal reactions, Accounts of Chemical Research 44 (2011) 666–676, https://doi.org/10.1021/ar200148z.
[51] D.P. Nair, M. Podgórski, S. Chatani, T. Gong, W. Xi, C.R. Fenoli, C.N. Bowman, The thiol-Michael addition click reaction: A powerful and widely used tool in materials chemistry, Chemistry of Materials 26 (2014) 724–744, https://doi.org/10.1021/cm402180t.
[52] K.G. Goswami, S. Mete, S.S. Chaudhury, P. Sar, E. Ksendzov, C.D. Mukhopadhyay, S.V. Kostjuk, P. De, Self-assembly of amphiphilic copolymers with sequence controlled alternating hydrophilic-hydrophobic pendant side chains, ACS Appl. Polym. Mater. 2 (2020) 2035–2045, https://doi.org/10.1021/acsapm.0c00204.
[53] A. Bera, D. Mukhopadhyay, K. Goswami, P. Ghosh, R. De, P. De, Fatty acid-based polymeric micelles to ameliorate amyloidogenic disorders, Biomaterials Science 10 (2022) 3466–3479, https://doi.org/10.1039/D2BM00359G.